



Analog IC Design

CMOS Technology

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Section 3 - Lab 04

Common drain amplifier

OP, AC and transient analysis

Device Sizing Using ADT

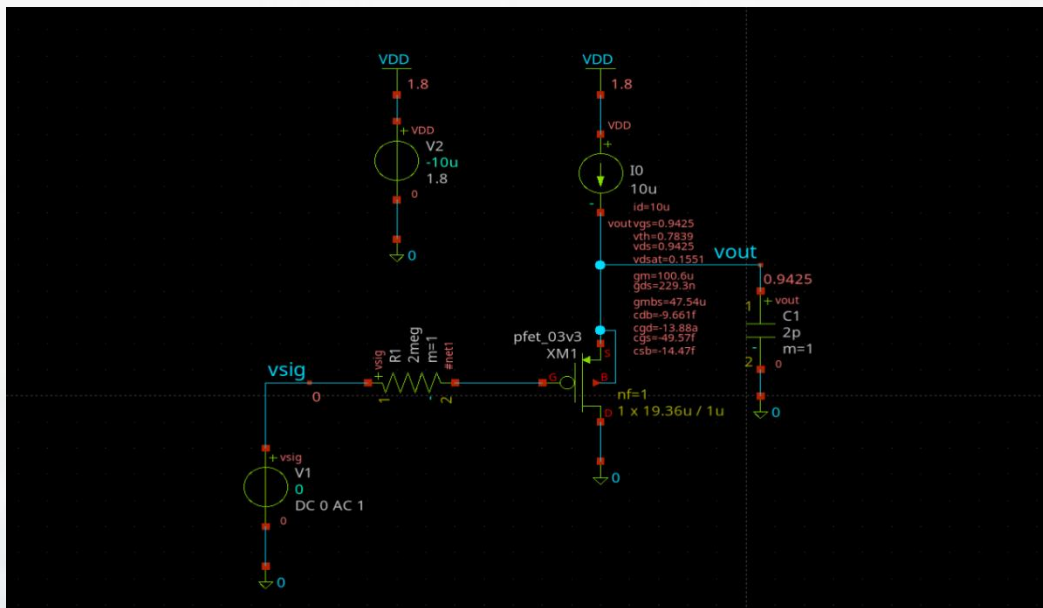
$$I_D = 10\mu A, \frac{g_m}{I_D} = 10\mu S/\mu A, L = 1\mu m, V_{DS} = V_{GS}.$$

$$\text{PMOS} \rightarrow V_{SB} = 0$$

$$\rightarrow W = 19.36\mu m$$

ID	10u	?
gm/ID	10	?
L	1u	?
VDS	VGS	?
VSB	0	?
Stack	1	?
Results:		
Name	TT-27.0	
4 W	19.36u	

OP Analysis



Note that the values of the capacitances are not accurate in xschem simulator:

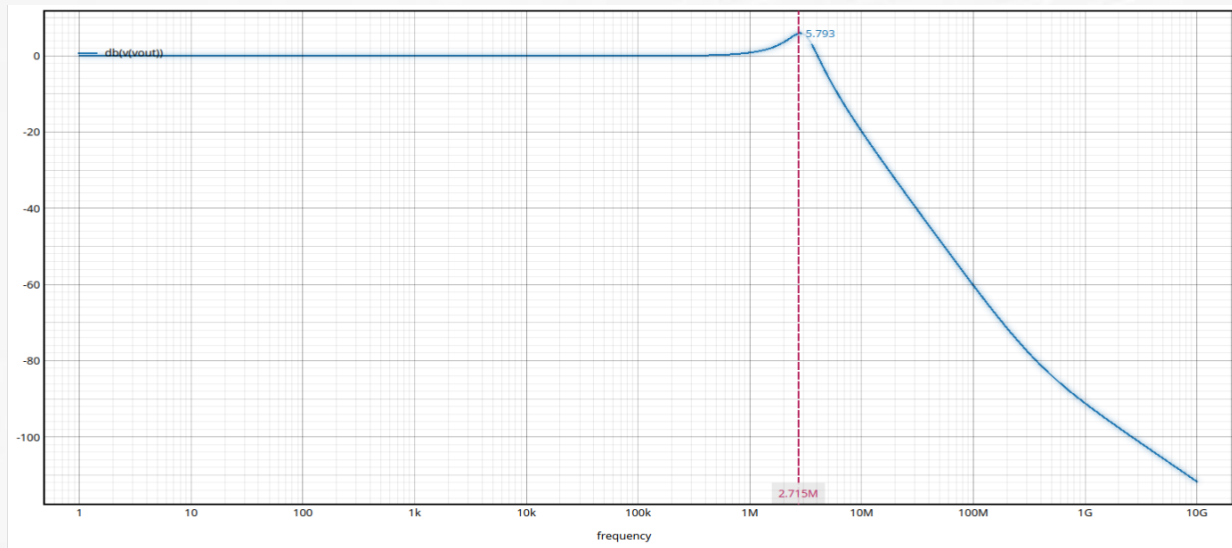
The capacitances we will use later, we can get them from ADT:

$$C_{gs} = 50fF, C_{sb} = 26.2fF, C_{gd} = 3fF$$

Check that the transistor operates in saturation.

$$I_D = 10\mu A \text{ and } V_{GD} = V_{GS} - V_{DS} = 0 < V_{TH} \text{ So } \rightarrow \text{the MOSFET in saturation.}$$

AC Analysis

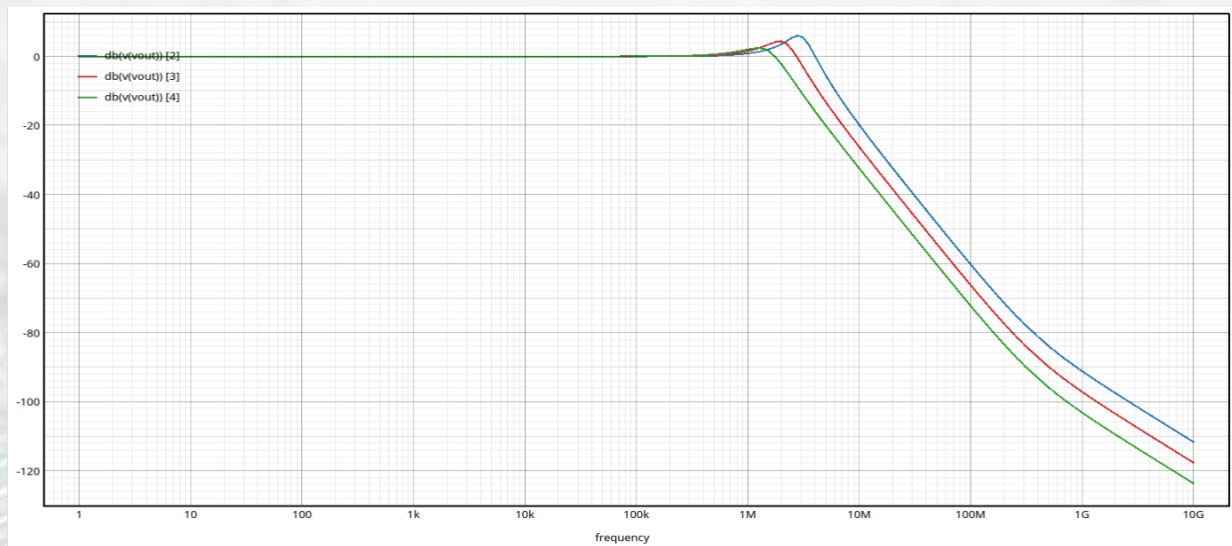


Bode plot: magnitude (dB)

There is a maximum peak at 2.72MHz around $5.8\text{dB} \approx 2v$.

Analytically calculate the quality factor (use approximate expressions). Is the system underdamped or overdamped?

since we drive large capacitive load $\rightarrow Q \approx \sqrt{\frac{g_m(C_{gs}+C_{gd})R_{sig}}{C_L}} \approx 2.3 > 0.5 \rightarrow \text{underdamped system}$



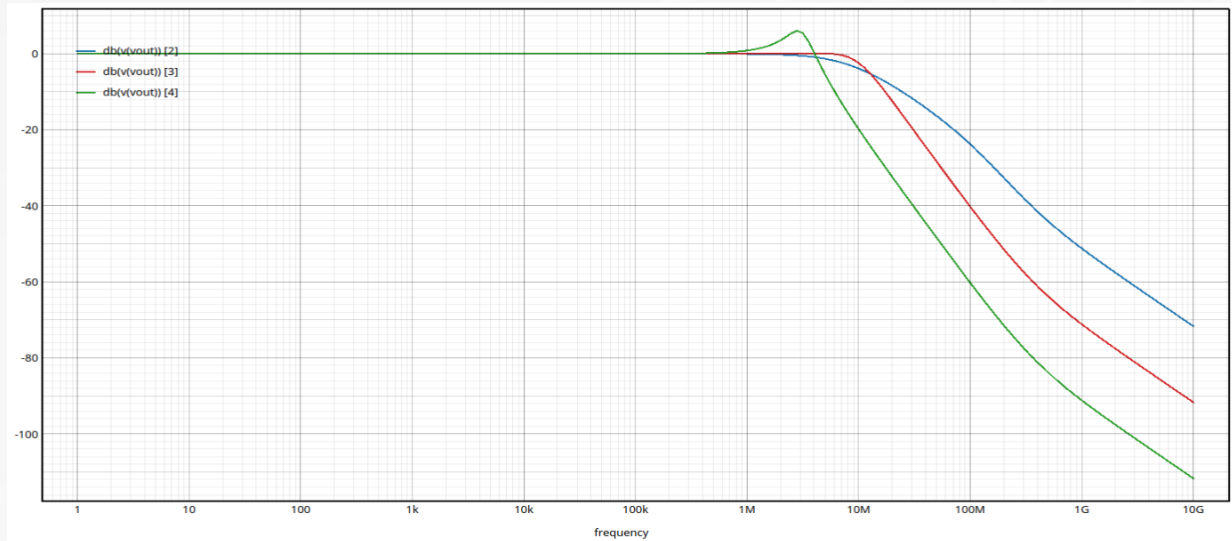
Bode plot: C sweeping [2pF, 4pF, 8pF]

Comment:

As C_L increases $\rightarrow Q$ decreases causing decreasing in the Gain peak as shown in the simulation results.

Also, the output pole decreases by increasing C_L causing decreasing in the BW.

```
CD Amplifier AC Sim Results
CL = 2p, Gain Peak (dB) = 5.9966
CL = 4p, Gain Peak (dB) = 4.33344
CL = 8p, Gain Peak (dB) = 2.3851
```



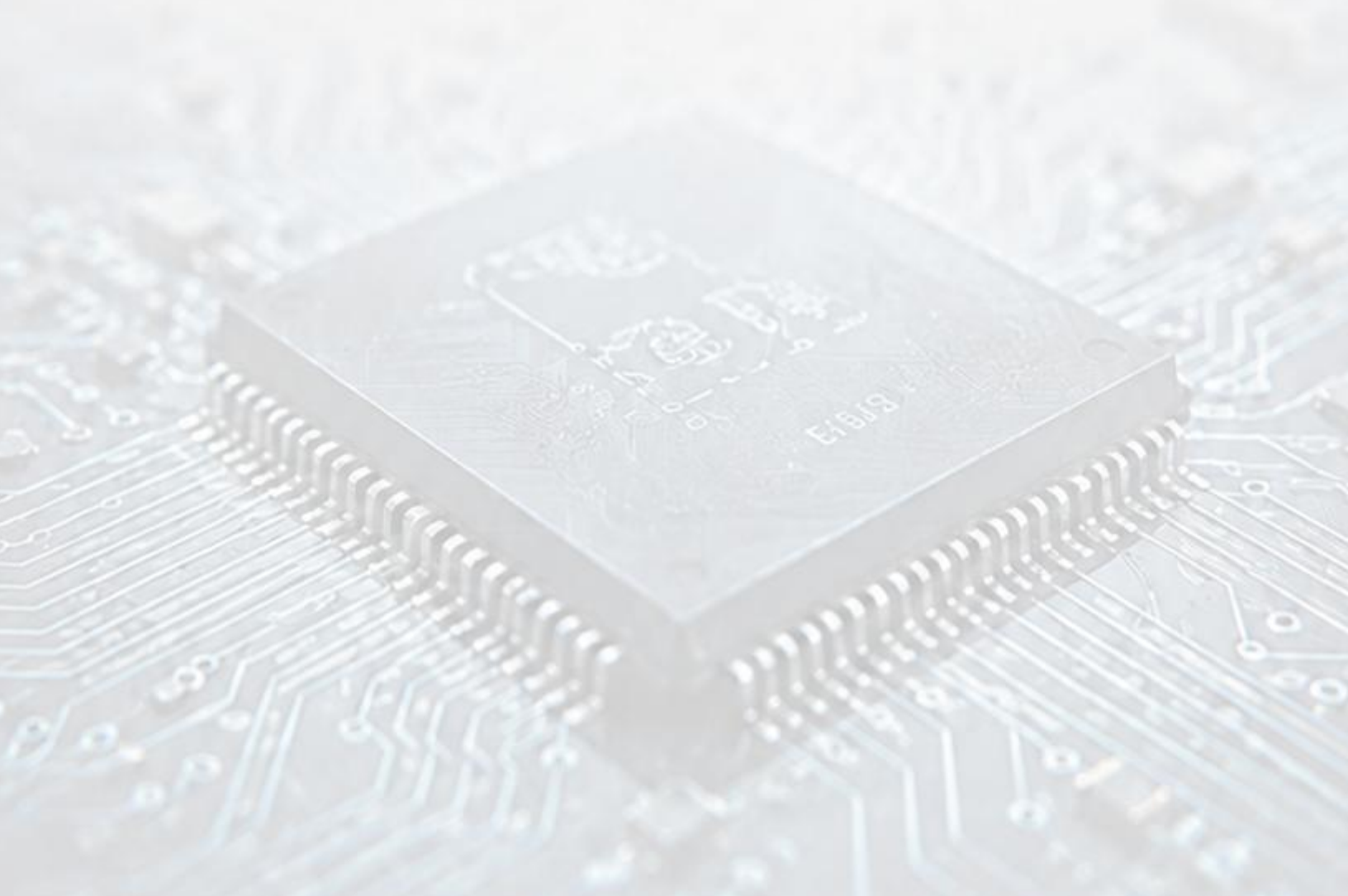
Bode plot: R_{sig} sweeping [20KΩ, 200 KΩ, 2MΩ]

Comment:

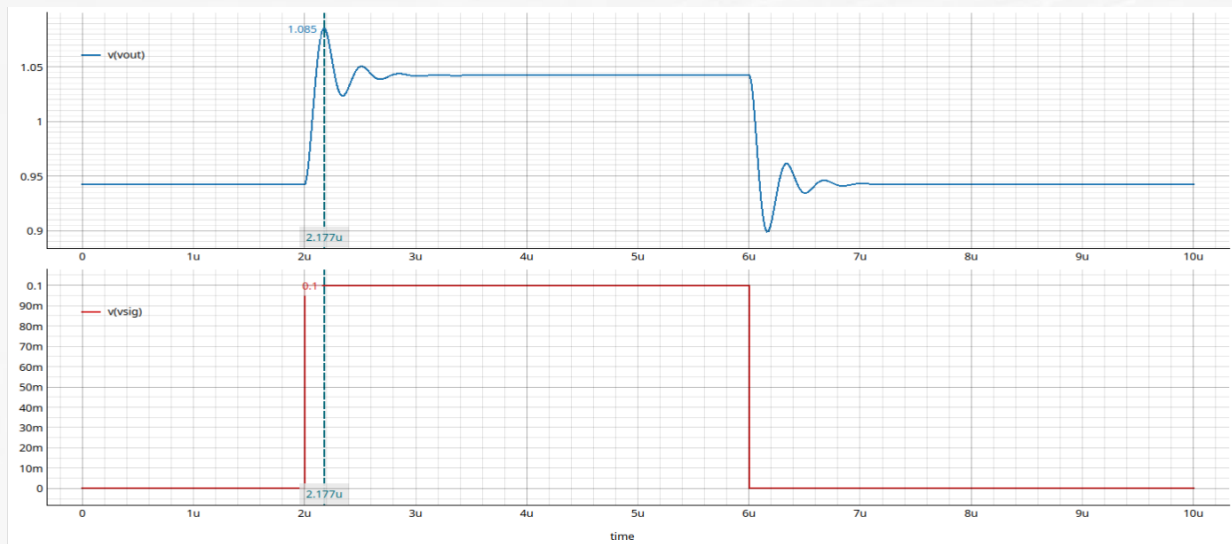
As R_{sig} increases $\rightarrow Q$ increases causing increasing in the Gain peak as shown in the simulation results.

Also, the output pole decreases by increasing R_{sig} causing decreasing in the BW.

```
RES = 20k, Gain Peak (dB) = -0.0197724
RES = 200k, Gain Peak (dB) = 0.121317
RES = 2meg, Gain Peak (dB) = 5.9966
```

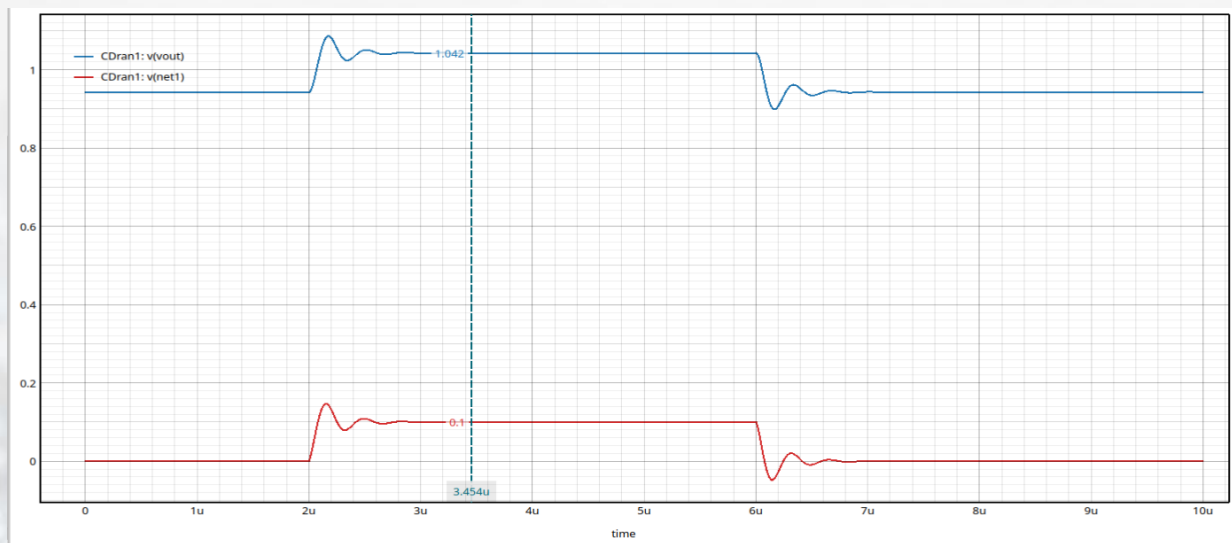


Transient Analysis



■ V_{sig} , ■ V_{out}

Calculate the DC voltage difference (DC shift) between V_{in} and V_{out} .



■ V_{in} , ■ V_{out}

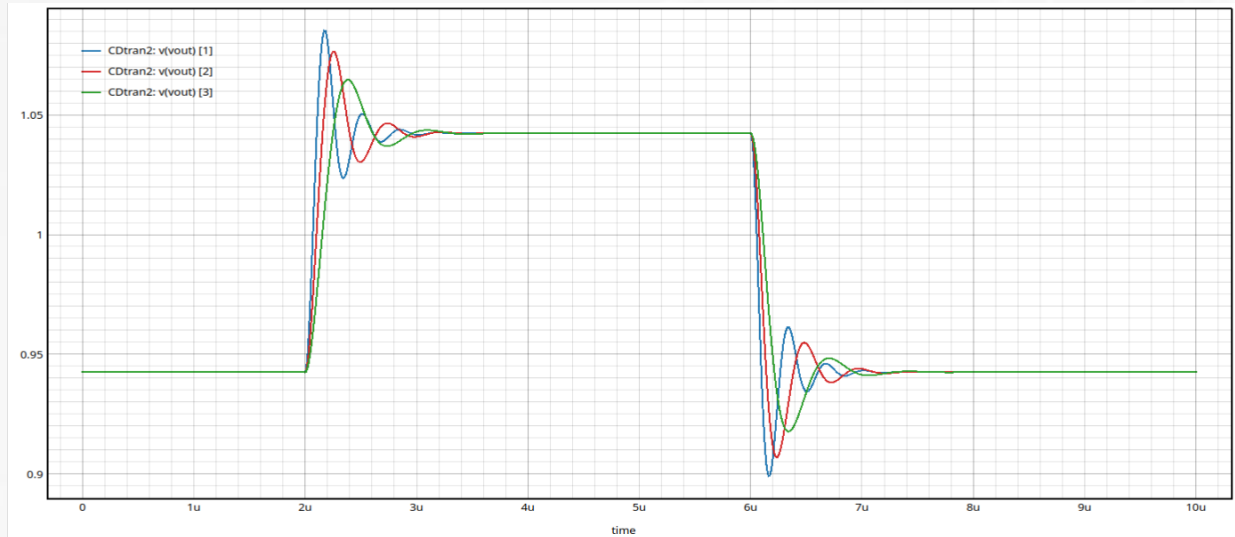
Comments:

The DC shift between V_{in} and $V_{out} \approx 1.05 - 0.1 = 0.95 \approx V_{GS}$

We can shift the signal down instead of shifting up by using NMOS technology.

There is ringing in the time domain response since $(Q = 2.3) > 0.707$.

The overshoot% = $100e^{-\frac{\pi}{\sqrt{4Q^2-1}}} = 100e^{-\frac{\pi}{\sqrt{4(2.3)^2-1}}} \approx 49.6\%$



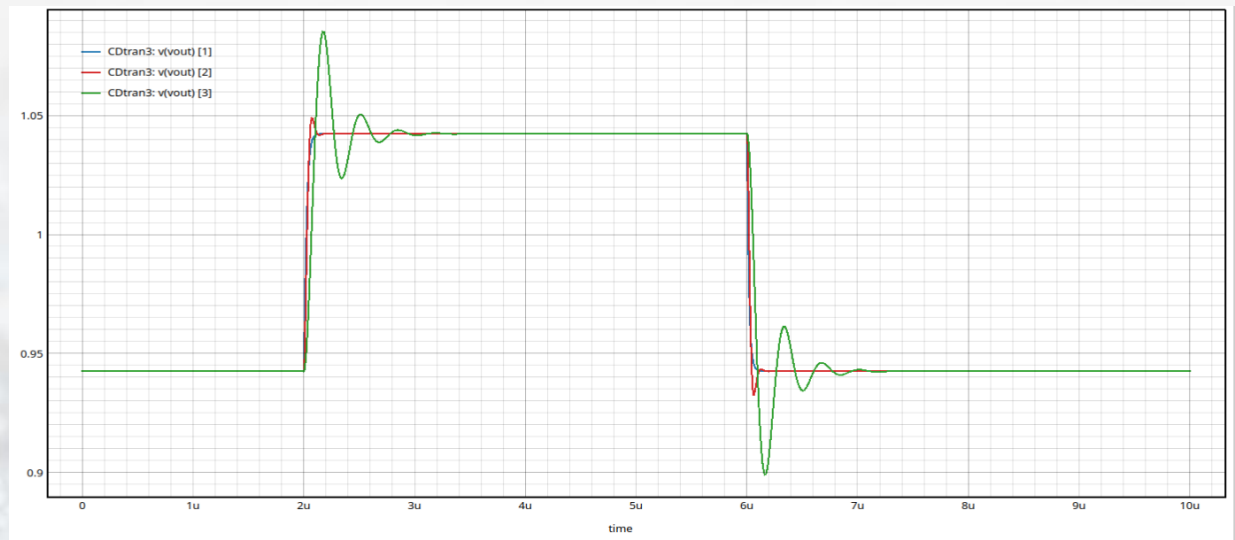
$V_{out}: C_L$ sweeping [2pF, 4pF, 8pF]

Comment:

As C_L increases $\rightarrow Q$ decreases causing decreasing in the overshoot as shown in the simulation results.

For all sweeping values for C_L , $Q > 0.5 \rightarrow$ under damped response.

```
CD Amplifier Transient Sim Results
CL = 2p, Overshoot (%) = 43.0232
CL = 4p, Overshoot (%) = 34.1077
CL = 8p, Overshoot (%) = 22.4211
```



$V_{out}: R_{sig}$ sweeping [20K Ω , 200 K Ω , 2M Ω]

Comment:

As R_{sig} increases $\rightarrow Q$ increases causing increasing in the overshoot as shown in the simulation results.

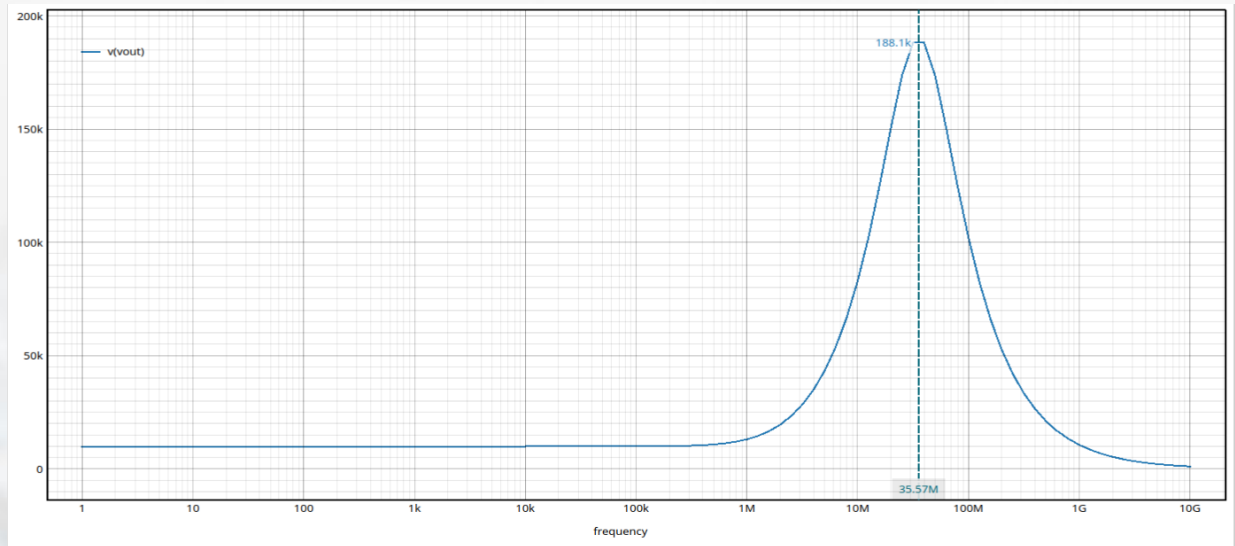
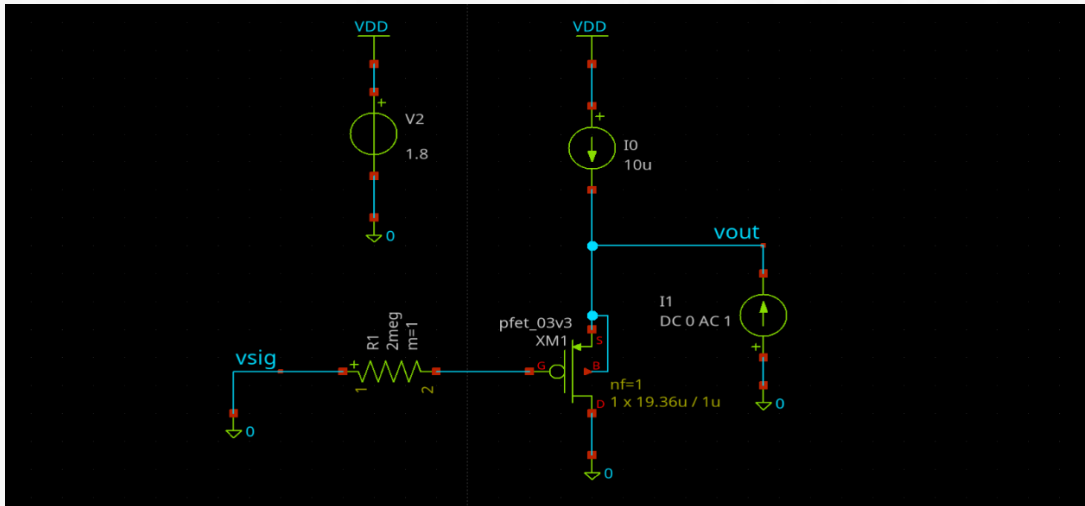
For $R = 20K\Omega \rightarrow Q = 0.235 < 0.5 \rightarrow$ over damped response.

For $R = 200K\Omega \rightarrow Q = 0.743 \sim 0.707 > 0.5 \rightarrow$ small under damped response.

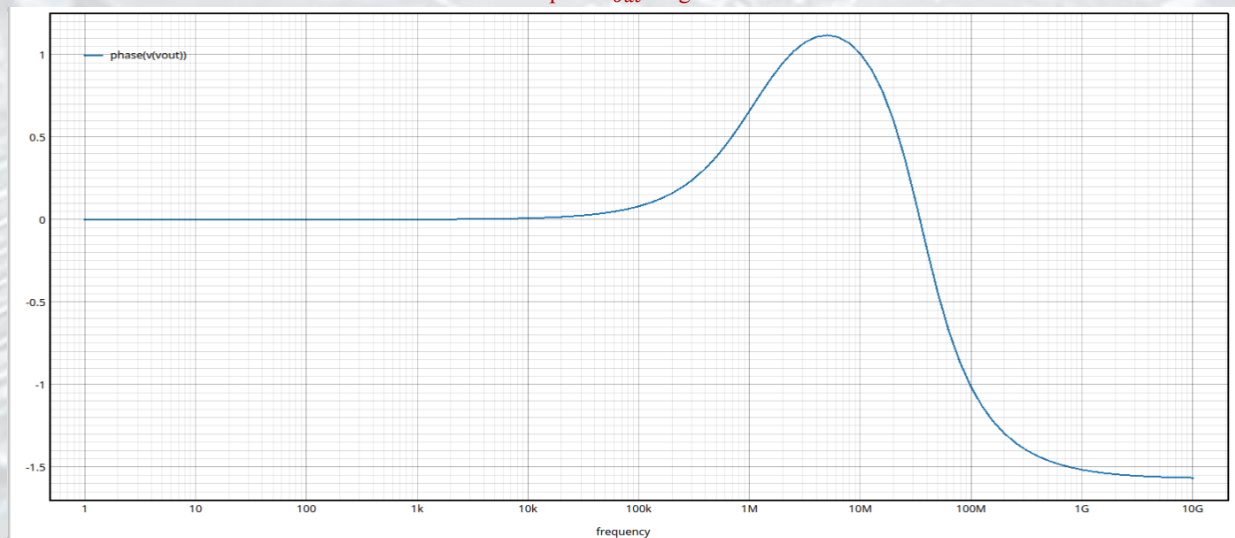
For $R = 2M\Omega \rightarrow Q = 2.35 \gg 0.5 \rightarrow$ under damped response.

```
Rsig = 20k, Overshoot (%) = 0
Rsig = 200k, Overshoot (%) = 6.58342
Rsig = 2meg, Overshoot (%) = 43.0232
```

ZOUT (inductive rise)



Bode plot: Z_{out} magnitude



Bode plot: Z_{out} phase

There is an inductive rise ($5MHz \rightarrow 35MHz$) due to effective inductance from the negative resistance.

Does Z_{out} fall at high frequency? Why?

There is another pole that appears at high frequency since the impedance of $C_{gd} \downarrow$ decreasing, $\omega \uparrow = \frac{1}{R_{sig}C_{gd} \downarrow}$ that will make Z_{out} falls at this frequency.

Analytically calculate the zeros, poles, and magnitude at low/high frequency for Z_{out} .

$$\omega_z \approx \frac{1}{R_{sig}C_{gs}} = \frac{1}{2\pi \cdot 2M\Omega \cdot 50fF} \approx 1.6MHz$$

$$\omega_{p1} \approx \frac{1}{\frac{1}{g_m}C_{gs}} = \frac{1}{2\pi \cdot \frac{1}{100\mu S} \cdot 50fF} \approx 318.3MHz$$

$$\omega_{p2} \approx \frac{1}{R_{sig}C_{gd}} = \frac{1}{2\pi \cdot 2M\Omega \cdot 3fF} \approx 26.5MHz$$

$$Z_{out} = \frac{1}{g_m} \times \frac{1+sR_{sig}C_{gs}}{(1+s\frac{C_{gs}}{g_m})(1+sR_{sig}C_{gd})}, \text{ note: } (1+sR_{sig}C_{gd}) \text{ is the 2}^{nd} \text{ pole.}$$

$$\text{At low frequency } Z_{out} \approx \frac{1}{g_m} = \frac{1}{100\mu S} = 10K\Omega$$

At high frequency $Z_{out} \approx R_{sig} = 2M\Omega$ ideally neglecting the effect of C_{gd} .

$$\text{Max peak: } Z_{out} \approx \frac{1}{g_m} \times \frac{1+sR_{sig}C_{gs}}{(1+sR_{sig}C_{gd})} \approx \frac{1}{g_m} \times \frac{C_{gs}}{C_{gd}} = \frac{1}{100\mu S} \times \frac{50fF}{3fF} \approx 166.2K\Omega \text{ analytical maximum peak.}$$

CD Amplifier Z_{out} Sim Results	
Z_{out} @ 1kHz	= 9917.68
Z_{out} peak (max)	= 188181

From the Bode plot, find the corresponding simulation (approximate values). Compare the results in a table.

parameter	Simulation	By hand
ω_z	2.5MHz	1.6MHz
ω_{p1}	60MHz	318.3MHz
ω_{p2}	26MHz	26.5MHz
Z_{out} low frequency	9.92K Ω	10K Ω
Z_{out} peak	188.2K Ω	166.2K

Comments:

At low frequency the output impedance approximately equals $\frac{1}{g_m}$

Ideally at high frequency the output impedance equals R_{sig}

But, since C_{gd} takes over at high frequency Z_{out} will fall down.

Since $R_{sig} \gg \frac{1}{g_m}$, C_{gd} pole comes before C_{gs} pole.

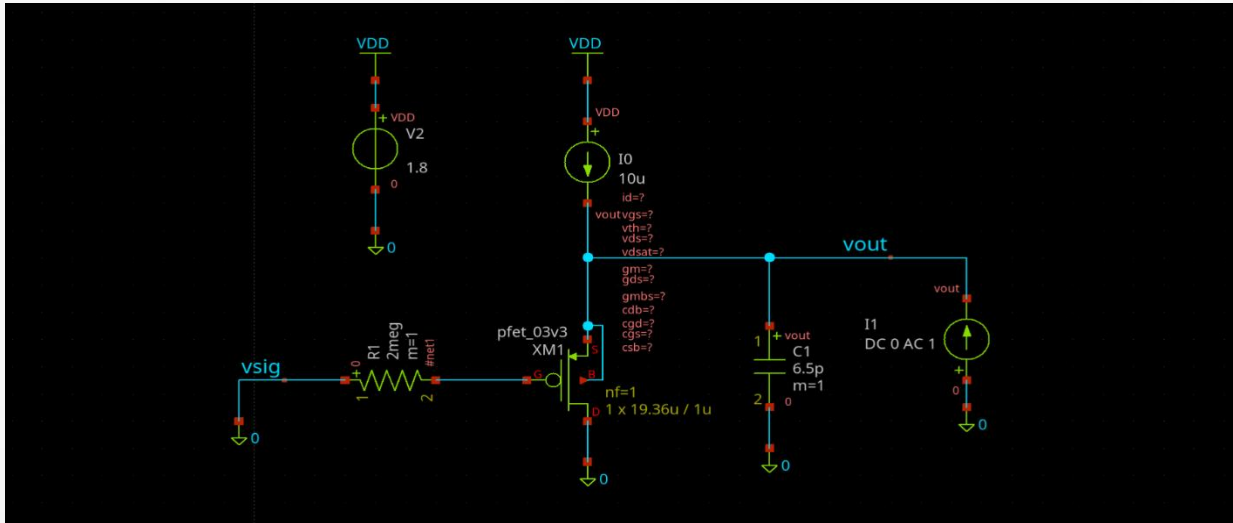
So that the actual peak will depend on the zero and C_{gd} pole, = 166.2K Ω

At very high frequency ($s \uparrow \uparrow$), $Z_{out} = \frac{1}{g_m} \times \frac{1+sR_{sig}C_{gs}}{(1+s\frac{C_{gs}}{g_m})(1+sR_{sig}C_{gd})} \approx \frac{1}{g_m} \times \frac{sR_{sig}C_{gs}}{(\frac{sC_{gs}}{g_m})(sR_{sig}C_{gd})} = \frac{1}{(s \uparrow \uparrow C_{gd})} \rightarrow 0$ as shown in the $|Z_{out}|$ curve at very high frequency.

Solving peaking and ringing problem

Simple solution

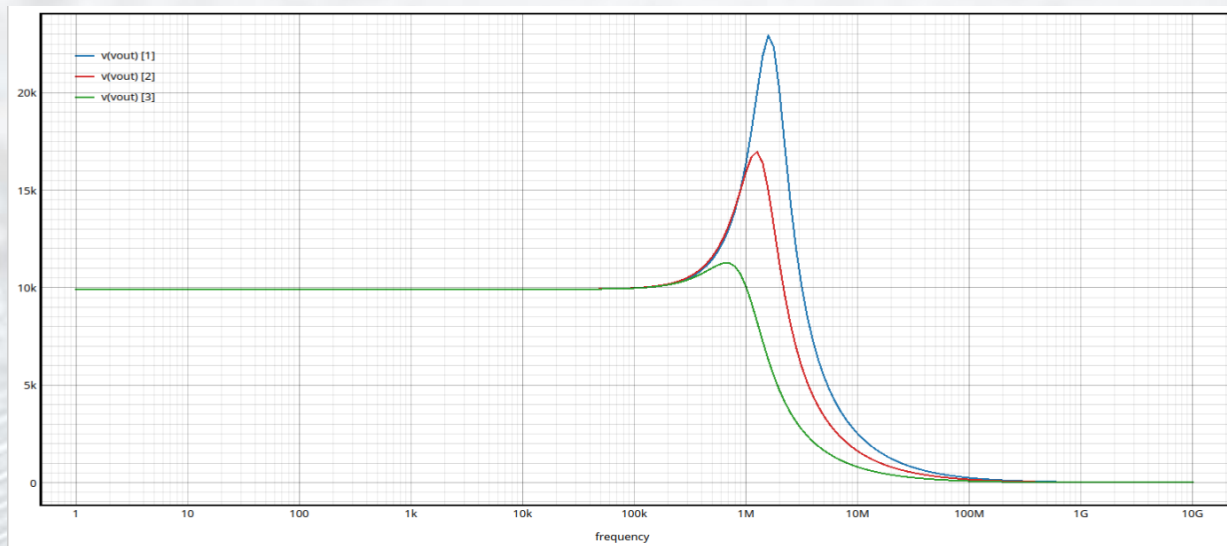
Place output pole away from input pole, (place it on the zero to cancel the inductive rise)



Assume C_{gs} bootstrapped:

$$\frac{1}{2\pi(R_S//R_{LFS})C_1}, R_S \rightarrow \infty, R_{LFS} = \frac{1}{g_m} \rightarrow \frac{1}{2\pi\left(\frac{1}{100\mu S}\right)C_1} = \omega_z = 2.4MHz \rightarrow C_1 \approx 6.5pF$$

But there will be some errors due to the approximations: C_1 must be larger.



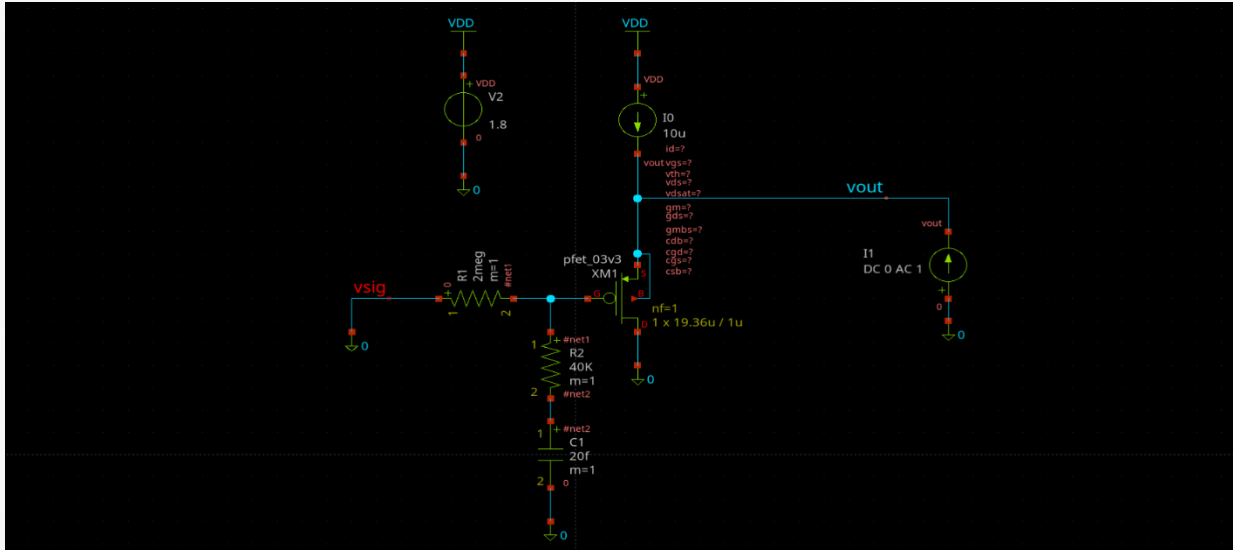
■ $C_L = 6.5pF$, ■ $C_L = 10pF$, ■ $C_L = 20pF$

Comment:

As C_L increases, the output pole moves to left reducing the peaking and also reducing the bandwidth.

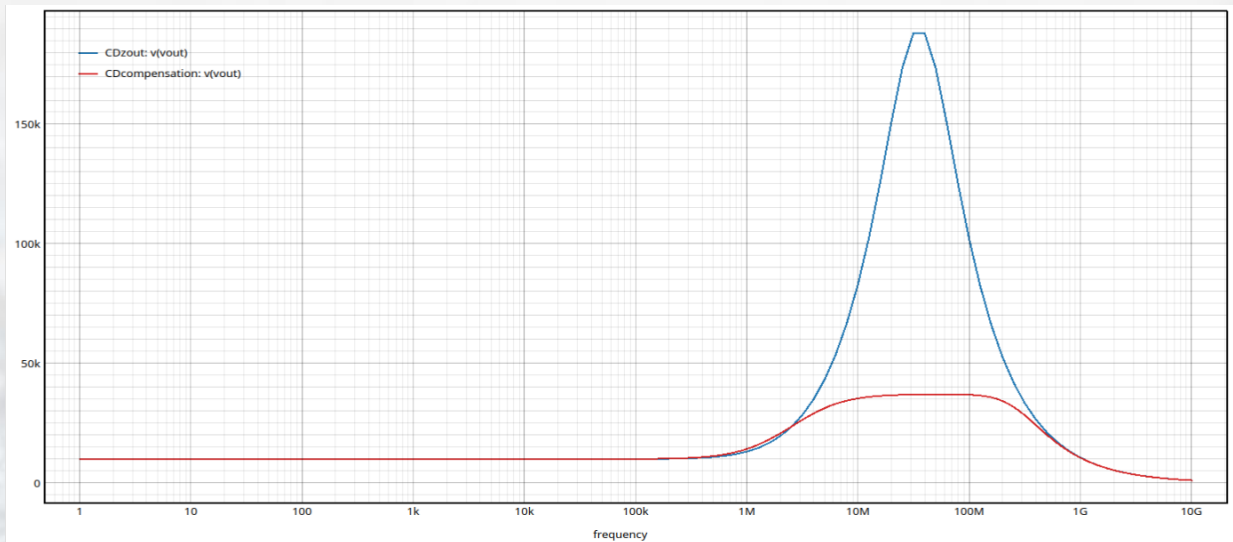
Smart solution: compensation network

Adding capacitive network to cancel the inductive rise



The expressions of C1 and R2 getting from [Johns and Martin, 2012] Section 4.4:

$$C_1 = \frac{g_m C_{gs} C_{sb}}{(g_m + C_{sb})(C_{gs} + C_{sb})} \approx 20fF, R_2 = \frac{(C_{gs} + C_{sb})^2}{g_m C_{gs} C_{sb}} \approx 40K\Omega$$



Z_{out} : ■ without compensation network, ■ with compensation network

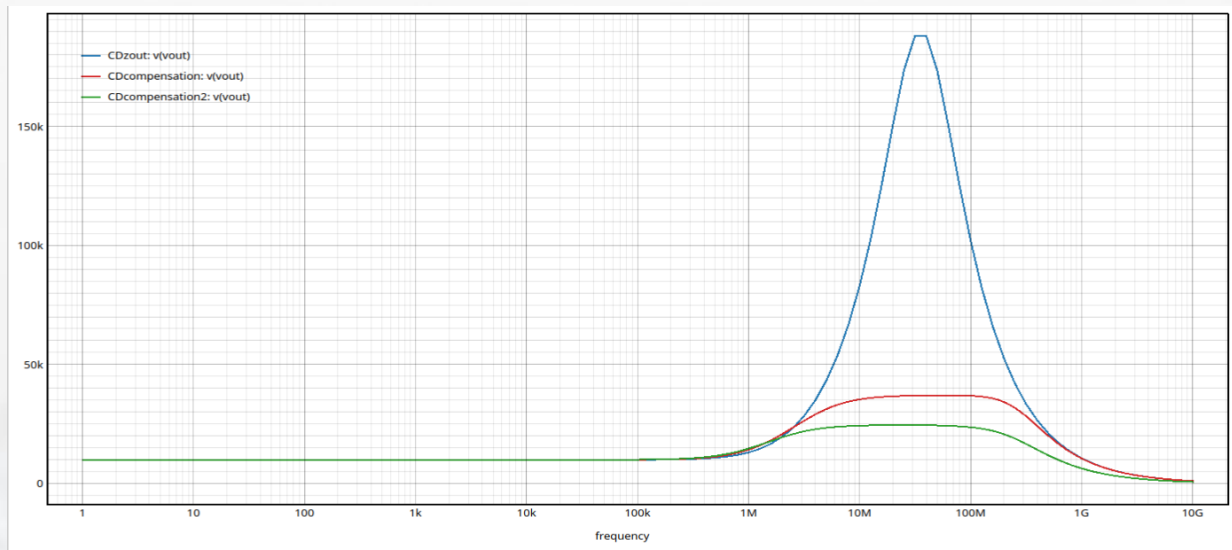
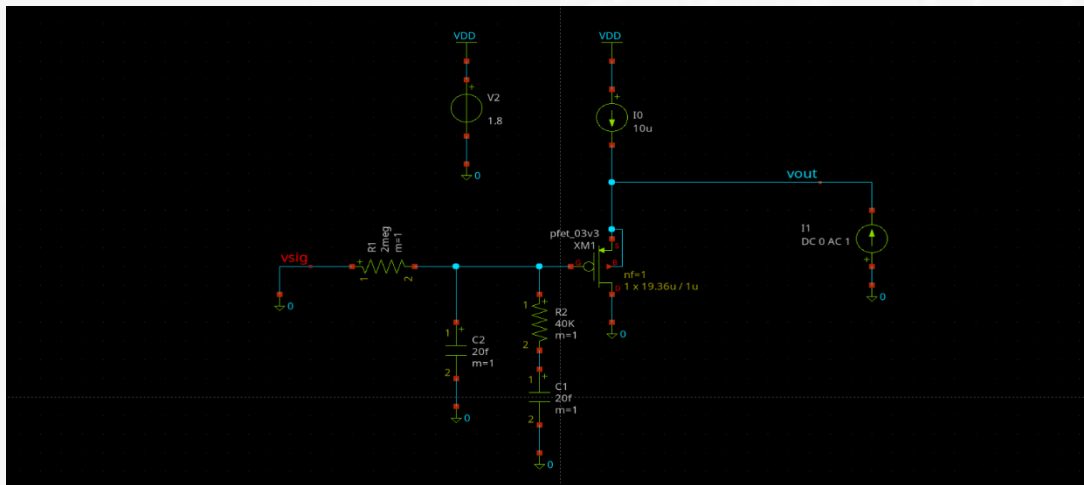
Comment:

The effect of the compensation network **significantly reduces the peaking** but there is still small peaking.

We can add parallel C2 to cancel the remaining peaking.

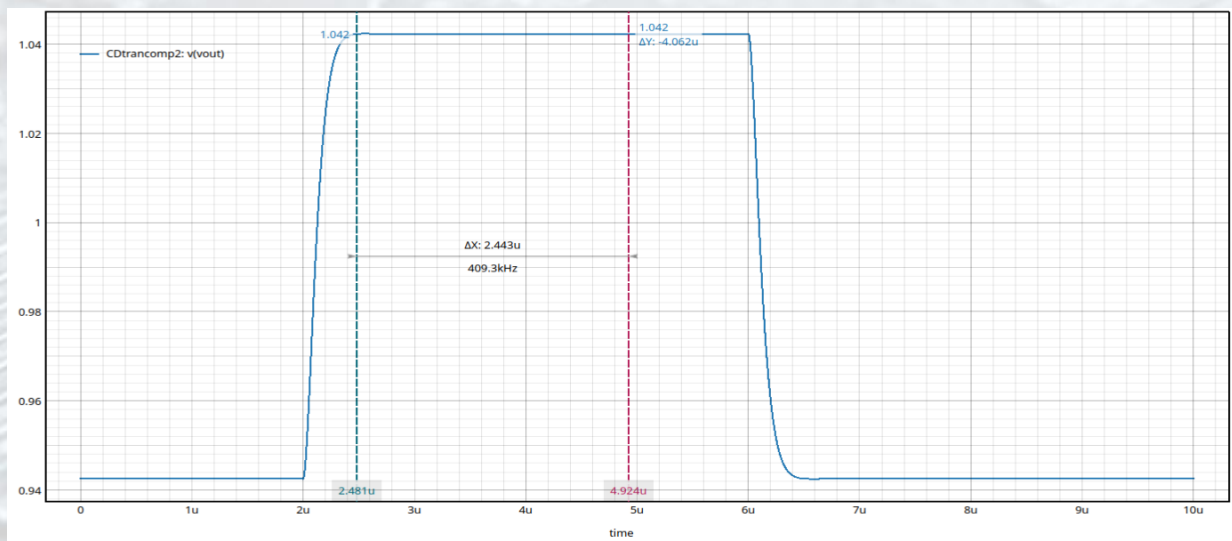
The expression of C2 getting from [Johns and Martin, 2012] Section 4.4:

$$C_2 = \frac{C_{gs} C_{sb}}{C_{gs} + C_{sb}} \approx 20fF$$



Z_{out} : ■ without compensation, ■ C_1, R_1 only, ■ C_1, C_2, R_1

Due to the approximations there will be **some peaking remaining** that will cause **overshoot in the transient** analysis, so we will need to increase the capacitances **by intuition** to some value: so i will set $C_1 = C_2 = 25fF$.



Transient analysis with compensation: no ringing nor overshoot